CLINICAL TEACHING NOTE | FAMILY MEDICINE

Type 2 Diabetes

Assessment and management in primary care

1. Diagnostic approach

Step 1: Classify and confirm the diagnosis (WHO/ADA criteria)

- Diagnostic thresholds: fasting plasma glucose ≥7.0 mmol/L (126 mg/dL), 2-hour OGTT glucose ≥11.1 mmol/L (200 mg/dL), HbA1c ≥48 mmol/mol (6.5%), or a random glucose ≥11.1 mmol/L with classic symptoms.
- Confirm with a repeat test on a separate day unless hyperglycaemia is unequivocal and the patient is symptomatic.
- Prediabetes: impaired fasting glucose (6.1–6.9 mmol/L), impaired glucose tolerance (2-hour OGTT 7.8–11.0 mmol/L), or HbA1c 42–47 mmol/mol (6.0–6.4%).
- Distinguish Type 2 diabetes from Type 1 diabetes, MODY, and secondary diabetes early, as this changes both urgency and treatment (see table).

Step 2: Key history

- Classic symptoms: polyuria, polydipsia, unexplained weight loss, fatigue, blurred vision, and recurrent infections such as candidiasis or UTIs.
- Risk factors: obesity or central adiposity, family history, gestational diabetes, ethnicity, sedentary lifestyle, hypertension, dyslipidaemia, and PCOS.
- Screen for complications at diagnosis and every review: visual symptoms, foot symptoms (numbness, pain, ulcers), erectile dysfunction, and symptoms of cardiovascular or renal disease.
- Medication history — corticosteroids, antipsychotics, and thiazides can precipitate or worsen hyperglycaemia.
- Features suggesting Type 1 diabetes or MODY: young age, lean body habitus, ketosis, or a strong multigenerational young-onset family history.

Step 3: Examination

- Weight, BMI, waist circumference, and blood pressure.
- Foot examination: pulses, monofilament or vibration sensation, skin integrity, and deformity, as part of an annual foot risk assessment.
- Fundoscopy, or referral for structured retinal screening.
- Skin: acanthosis nigricans as a marker of insulin resistance, and review of injection sites if the patient is on insulin.
- Signs of associated conditions: PCOS features, or signs of another endocrinopathy if secondary diabetes is suspected.

2. Distinguishing the common causes

	Type 2 diabetes	Type 1 diabetes	MODY	Secondary diabetes
Typical onset	Usually over 40, gradual (increasingly seen at younger ages)	Any age, often childhood or young adulthood; can present in adulthood as LADA	Usually under 25, with a strong autosomal dominant family history	Any age, temporally related to the underlying cause

	Type 2 diabetes	Type 1 diabetes	MODY	Secondary diabetes
Body habitus	Often overweight or obese	Often normal weight or lean	Normal weight	Variable
Key features	Insulin resistance, metabolic syndrome features, gradual symptom onset	Rapid onset, weight loss, ketosis-prone, may present with DKA	Mild, often stable hyperglycaemia (varies by subtype); strong multigenerational family history	Pancreatic disease, endocrinopathy (e.g. Cushing's, acromegaly), or drug-induced (steroids, antipsychotics)
Autoantibodies / genetics	Negative; C-peptide preserved	GAD, IA-2, ZnT8 positive; low C-peptide	Negative; specific gene mutation (e.g. HNF1A, GCK)	Negative
Initial management	Lifestyle measures plus metformin	Insulin required from diagnosis	Depends on subtype — some respond to sulfonylureas, others need no treatment	Treat the underlying cause plus glucose-lowering therapy as needed

3. Investigations

A reasonable primary-care set at diagnosis and for ongoing monitoring:

- HbA1c at diagnosis and every 3–6 months depending on control and treatment changes.
- Fasting lipid profile and renal function (eGFR, urine albumin:creatinine ratio) at diagnosis and annually.
- LFTs at baseline (and before or during certain medications), and TSH if clinically indicated.
- Urinalysis for glucose and ketones; consider C-peptide and autoantibodies if Type 1 diabetes or MODY is suspected.
- Annual retinal photography screening and a foot risk assessment.
- Cardiovascular risk assessment — blood pressure, lipids, and smoking status — at diagnosis and at regular review.

Red flags: same-day or emergency referral

- Symptoms or signs of diabetic ketoacidosis: vomiting, abdominal pain, Kussmaul breathing, drowsiness, or ketones on urinalysis — this is an emergency.
- Hyperosmolar hyperglycaemic state: marked hyperglycaemia, dehydration, and altered consciousness in Type 2 diabetes — an emergency.
- Suspected new-onset Type 1 diabetes (rapid onset, weight loss, ketosis) — refer the same day for insulin initiation.
- Foot ulceration with signs of infection or ischaemia — urgent referral via the diabetic foot pathway.
- Severe hypoglycaemia, especially if recurrent or the patient has impaired awareness.
- Sudden visual loss or signs of proliferative retinopathy or vitreous haemorrhage.

4. Management

General measures for all causes

- Provide structured education on diabetes, diet, and self-monitoring at diagnosis and periodically thereafter.
- Lifestyle measures: weight management, dietary modification (reduced refined carbohydrate and sugar, increased fibre), regular physical activity, and smoking cessation.

- Individualise the HbA1c target based on age, comorbidity, hypoglycaemia risk, and life expectancy — commonly 48–58 mmol/mol (6.5–7.5%), relaxed in frailty.
- Address cardiovascular risk reduction: blood pressure control, statin therapy per risk assessment, and aspirin only if there is an additional indication.

Glucose-lowering therapy

Line	Drug	Typical dose	Key cautions
First line	Metformin	Start 500 mg once or twice daily with food, titrating to 500–1000 mg twice daily (max around 2 g/day)	GI side effects; avoid or reduce dose in significant renal impairment; stop temporarily during acute illness or contrast imaging.
Second line	SGLT2 inhibitor (e.g. dapagliflozin, empagliflozin)	Standard once-daily dose	Genital mycotic infections, euglycaemic DKA risk, volume depletion; offers additional cardiorenal benefit.
Second line	GLP-1 receptor agonist (e.g. semaglutide, liraglutide)	Titrated per product, weekly or daily injection	GI side effects, rare pancreatitis; associated with weight loss and cardiovascular benefit in high-risk patients.
Second line	DPP-4 inhibitor (e.g. sitagliptin)	Standard once-daily dose	Generally weight-neutral and well tolerated; adjust for renal function.
Second line	Sulfonylurea (e.g. gliclazide)	Start low and titrate	Hypoglycaemia risk and weight gain; use cautiously in the elderly and in renal impairment.
Third line / insulin	Basal insulin (e.g. insulin glargine)	Start weight-based, titrate to fasting glucose target	Hypoglycaemia risk, weight gain; requires injection technique and storage education.

Choice of second-line agent

- Prefer an SGLT2 inhibitor or GLP-1 receptor agonist with proven cardiovascular or renal benefit in patients with established cardiovascular disease, heart failure, or chronic kidney disease.
- Weigh weight, hypoglycaemia risk, cost or access, and patient preference when selecting an agent.

Insulin initiation

- Consider when HbA1c remains significantly above target despite optimised oral or injectable therapy, or with symptomatic hyperglycaemia or contraindications to other agents.
- Start with basal insulin in most patients with Type 2 diabetes and involve a diabetes specialist nurse or dietitian for education.

Hypoglycaemia management (if on insulin or a sulfonylurea)

- Educate the patient on recognition and immediate treatment: fast-acting carbohydrate, retesting, then a longer-acting carbohydrate.
- Review the regimen and doses after any significant hypoglycaemic episode.

5. Follow-up

- Check HbA1c every 3–6 months until stable at target, then at least 6-monthly.
- Annual review: weight/BMI, blood pressure, foot examination, renal function and urine ACR, lipid profile, and retinal screening.
- Review medication at every visit, particularly renal function-dependent dosing and hypoglycaemia risk with advancing age.

- Reassess and individualise targets as comorbidity, frailty, or life expectancy change over time.

Key take-home points

- Confirm the diagnosis with a repeat test unless hyperglycaemia is unequivocal and symptomatic.
- Distinguish Type 2 diabetes from Type 1, MODY, and secondary diabetes early — this changes urgency and treatment.
- Metformin remains first line; choose second-line agents based on cardiovascular and renal risk, not HbA1c alone.
- Individualise HbA1c targets, especially in older or frail patients, to avoid harmful hypoglycaemia.
- Annual complication screening — eyes, feet, kidneys, cardiovascular risk — is part of routine management, not optional.

References

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4. International Diabetes Federation. IDF Clinical Practice Recommendations for Managing Type 2 Diabetes in Primary Care.

5. Holt RIG, DeVries JH, Hess-Fischl A, et al. The Management of Type 1 Diabetes in Adults: A Consensus Report by ADA and EASD. Diabetes Care. 2021;44(11):2589-2625.

6. World Health Organization. Classification of Diabetes Mellitus, 2019.

This summary is an educational aid compiled from standard family medicine and specialty references; it is not a substitute for local clinical guidelines or the primary literature.